

MD. ROFAZ HASAN RAFIU

AI Engineer • Generative AI & Computer Vision • Backend Systems

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PROFESSIONAL SUMMARY

Computer Science & Engineering student at Rajshahi University of Engineering & Technology (RUET) building AI-powered software systems across Generative AI, computer vision, and backend engineering. Experienced in integrating local language models, structured AI workflows, client-side computer vision, REST APIs, PostgreSQL, and real-time systems. Focused on developing reliable AI applications with practical fallbacks, validation, and production-oriented software architecture.

EDUCATION

Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering 2023 – 2027

- Admitted through national combined engineering university admission process (CUET, KUET, RUET).
- Relevant Coursework: Linear Algebra, Probability & Statistics, Algorithms & Data Structures, Database Systems, Discrete Mathematics.

St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science 2022

- GPA: 5.00 / 5.00.

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, Generative AI, Large Language Models (LLMs), Local SLM Deployment (Ollama, Qwen 2.5), Model Inference, Structured Outputs, Prompt Engineering, Deterministic Fallbacks, AI Evaluation.

Computer Vision: OpenCV.js (WebAssembly), TensorFlow.js (BlazeFace, WebGL), Image Preprocessing (Adaptive Thresholding, Canny Edge Detection, 4-Point Homography Warp, Contour Analysis, ROI Segmentation), Pixel Intensity Analysis.

Programming & Frameworks: Python (FastAPI, NumPy, OpenCV basics), TypeScript, Go 1.22+, C++, React 19, Next.js 15, Flutter (Dart).

Backend, Data & Infrastructure: PostgreSQL (Prisma ORM, Indexing, Connection Pooling), Redis, SQLite, Docker, Linux/Bash, WebSockets, RESTful APIs, Edge CDN Caching, Git, GitHub Actions.

SELECTED AI & SOFTWARE ENGINEERING PROJECTS

Digital School — AI Systems Engineer [Next.js 15, TypeScript, PostgreSQL, Prisma, Ollama, Qwen 2.5, TensorFlow.js] 2024 – Present

Full-stack education platform combining automated assessment, diagnostic analytics, computer vision, and backend services. Live: <https://rofazacademy.dev>

- Local SLM Inference:** Integrated Qwen 2.5 via Ollama to classify student exam errors across structured academic concept categories using constrained prompts and structured schema outputs.
- Deterministic Resilience:** Implemented a PostgreSQL-based deterministic fallback for diagnostic reporting when local model inference is unavailable or saturated, ensuring continuous service uptime.
- Client-Side Computer Vision:** Developed browser-based computer vision workflows using TensorFlow.js and WebGL for real-time examinee presence tracking without server video streaming overhead.
- Data Infrastructure & Caching:** Designed multi-tenant PostgreSQL schemas with Prisma ORM and configured edge CDN caching headers, optimizing connection pooling for concurrent assessment workloads.

OMRView — Computer Vision Engineer [TypeScript, React 19, Vite, OpenCV.js WebAssembly] 2025

Client-side optical mark recognition and document geometry rectification engine. Code: github.com/rofazhasan/OMRView

- WebAssembly Processing:** Moved document-processing operations to client-side OpenCV.js/WebAssembly, executing image rectification and mark recognition directly in the browser.
- Geometric Rectification:** Implemented a multi-stage vision pipeline with adaptive thresholding, Canny edge detection, and 4-point homography perspective warps to rectify skewed camera captures.
- Mark Classification:** Constructed a bubble fill classifier evaluating pixel intensity histograms across segmented Regions of Interest (ROIs) against baseline illumination to detect filled marks and multi-fill collisions.

FocusGuard — Systems Software Engineer [Go 1.22, Docker, WebSockets, Chrome MV3, SQLite] 2025

Cross-platform daemon and browser extension for real-time digital policy enforcement. Code: github.com/rofazhasan/FocusGuard

- Distributed Daemon:** Built a centralized Go 1.22 service coordinating real-time focus session state across desktop and mobile clients using persistent WebSockets.
- Browser-Level Policy:** Implemented declarative browser-level network filtering rules using Chrome's Declarative Net Request API, avoiding per-request JavaScript interception overhead.
- Local-First Synchronization:** Engineered local SQLite policy caching with delta synchronization and integrated an Android VPN UDP DNS sinkhole intercepting restricted domains.

Atlas — macOS Systems & Financial OS Architect [Swift 6, SwiftUI, SwiftData, Apple Vision OCR, CryptoKit, AppKit] 2024

Native desktop operating system for financial tracking, cryptographic security, and document OCR. Code: github.com/rofazhasan/atlas

- Cryptographic Vault:** Implemented zero-knowledge AES-256 GCM payload encryption with PBKDF2 key derivation and Touch ID / LocalAuthentication biometric gatekeeping.
- Vision OCR Pipeline:** Integrated Apple Vision framework for offline document scanning, parsing tabular expense records directly into native SwiftData models.
- Native Packaging:** Engineered automated DMG build tooling with ICNS icon compilation, code-signing configuration, and release binary packaging for Apple Silicon.

Krishi Bondhu AI — Mobile & Edge Engineer [Flutter, Dart, Riverpod, SQLite, Camera Hardware] 2024

Offline-first mobile crop health assistant for field image collection and diagnosis. Code: github.com/rofazhasan/Krishi_Bondhu_AI

- Developed an on-device camera image acquisition pipeline with standardized leaf capture, local SQLite persistence, and bilingual audio feedback to support low-connectivity agricultural field use.

Additional Open-Source Systems & Repositories

GitHub Profile: github.com/rofazhasan

- **BioEntry** (github.com/rofazhasan/BioEntry): Biometric access verification and attendance logging engine with tamper-resistant audit records.
- **sdms** (github.com/rofazhasan/sdms): High-concurrency Student Data Management System with relational schemas and role-based permissions in TypeScript.
- **Meal-Management** (github.com/rofazhasan/Meal-Management): Fullstack automated resource ledger and meal allocation system built with TypeScript and modern web standards.
- **study-companion** (github.com/rofazhasan/study-companion): Cross-platform academic workflow and habit tracking mobile application built with Flutter, Dart, Kotlin, and C++.
- **Election_Game** (github.com/rofazhasan/Election_Game): Real-time political simulation engine and state machine architecture engineered in Godot Engine / GDScript.

HONORS & LEADERSHIP

Creative Talent Hunt — Regional Champion, Mathematics & Computing

Government of Bangladesh • 2018

Recognized by the Government of Bangladesh for demonstrated excellence in mathematics, analytical reasoning, and computing.

RUET CSE Peer Workshop Instructor & Competitive Programmer

2023 – Present

Organized and taught hands-on peer workshops on Data Structures, Algorithms, and System Architecture; active participant in programming contests.